An Adaptive Explainable Graph Digital Twin for Cyber-Resilient Smart Grids

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***Abstract*—Supervisory Control and Data Acquisition (SCADA) telemetry in modern power grids remains vulnerable to False Data Injection Attacks (FDIAs) engineered to bypass classical Weighted Least Squares (WLS) state estimation filters. Standard machine learning detectors rely on fixed input dimensions and collapse under topology reconfigurations such as line switching transients or breaker outages. This study presents a topology-invariant Graph Digital Twin (GDT) framework linking physical grid dynamics with an architectural multi-task GraphSAGE neural network. By operating directly on spatial admittance graphs of SCADA measurements, the model concurrently executes grid-level attack detection and substation-level anomaly localization within a single forward pass. Feature attributions computed via Integrated Gradients (IG) identify compromised sensor channels without relying on perturbation heuristics. Validated on the IEEE 14-bus test feeder under non-linear AC power flow constraints, the GraphSAGE backbone achieves a nominal detection F1-score of $98.25\% \pm 0.34\%$ and localization F1-score of $97.62\% \pm 0.84\%$ across five random initialization seeds. Under zero-shot transmission line outages, the GDT maintains $92.80\% \pm 3.37\%$ detection and $86.96\% \pm 1.52\%$ localization F1-scores, outperforming conventional non-relational Multi-Layer Perceptrons. Furthermore, an integrated active healing mechanism substitutes compromised telemetry with baseline load forecasting profiles, reducing state tracking mean squared error (MSE) from 51.19 MW^2 under active attack to 21.91 MW^2 (57.21% recovery).**

***Index Terms*—Graph Digital Twin, Cyber-Physical Power Systems, GraphSAGE, Explainable AI, False Data Injection Attacks, Closed-Loop Grid Healing.**

I. Introduction

Modern power transmission grids depend on wide-area Supervisory Control and Data Acquisition (SCADA) systems and Remote Terminal Units (RTUs) to supply state estimation (SE) routines with bus voltage magnitudes, phase angles, and active/reactive power flows. While digital communication links enable real-time state awareness and dynamic generation dispatch, they expose physical transmission assets to coordinated cyber-physical tampering. Adversaries possessing partial knowledge of the network admittance matrix Y_{bus} construct stealthy False Data Injection Attacks (FDIAs) [3], [4]. By injecting structured measurement corruptions that satisfy Kirchhoff's current and voltage laws, these attacks alter estimated grid states while maintaining bad data detection (BDD) residuals below statistical χ^2 rejection thresholds.

Conventional machine learning approaches, including Multi-Layer Perceptrons (MLPs) and Convolutional Neural Networks (CNNs), have been applied to recognize anomalous SCADA patterns [5], [6]. However, flat tabular classifiers bind feature representations to a static input matrix ordering. In utility operations, transmission line trips, transformer maintenance, and bus-bar reconfigurations alter the spatial connectivity and measurement layout. When grid topology evolves, fixed-matrix MLPs experience spatial dimension mismatches or invalid correlation assumptions, degrading classification accuracy unless completely retrained.

Graph Neural Networks (GNNs) resolve topological constraints by mapping electrical networks directly to non-Euclidean graph structures where substations form nodes and transmission branches form edges [2], [12], [14]. Despite this spatial adaptivity, standard isotropic graph filters such as Graph Convolutional Networks (GCNs) aggregate neighborhood representations using uniform averaging. In deep architectures or dense physical clusters, uniform aggregation induces *over-smoothing*, where node representations converge toward identical mean vectors, obscuring localized substation-level anomaly signatures. Additionally, most existing GNN anomaly detectors function as open-loop alarm systems, lacking deterministic feature attribution mechanisms or active feedback control loops required to isolate corrupted telemetry and restore physical grid tracking.

To resolve these operational challenges, we formulate a cyber-resilient Graph Digital Twin (GDT) architecture that establishes continuous, bidirectional synchronization between the physical power grid and a dynamic virtual replica. The virtual twin maps incoming SCADA telemetry onto a spatial admittance graph and deploys a multi-task GraphSAGE backbone to perform joint attack classification and node-level localization in a single pass. Rather than relying on stochastic perturbation explainers such as SHAP or LIME, we integrate gradient path integration via Integrated Gradients (IG) to calculate sensor-level attributions satisfying axiomatic completeness and symmetry bounds. Flagged substations trigger a closed-loop healing module that isolates corrupted RTU telemetry and substitutes forecasted baseline loads to preserve state tracking stability.

A. Research Gap

Prior studies evaluating GNNs in smart grid anomaly detection primarily assume invariant network topologies and restrict

evaluation to binary classification. Two primary technical gaps remain:

1) Isotropic graph convolutions blur node-specific anomaly signals through over-smoothing, preventing fine-grained substation localization during targeted cyber-physical attacks.
2) Open-loop detection frameworks fail to deliver axiomatic sensor attributions or dynamic state reconstruction mechanisms necessary for automated telemetry isolation and closed-loop grid recovery.

This work resolves both gaps by designing a localization-aware GraphSAGE backbone coupled with an axiomatic attribution engine and a closed-loop state reconstruction control loop.

B. Main Contributions

The key contributions of this paper are summarized as follows:

- **Cyber-Physical GDT Architecture**: We design a bidirectional Digital Twin coupling that continuously parses SCADA measurement vectors into spatial graph representations for joint FDIA detection and substation localization.
- **Concatenation-Based GraphSAGE Backbone**: We employ a concatenation-based GraphSAGE convolution scheme ($[h_v^{(k-1)} \parallel h_{\mathcal{N}(v)}^{(k)}]$). By preserving target node feature signatures alongside neighborhood aggregations, the model resists over-smoothing and maintains high node-level localization precision.
- **Axiomatic Explainability & Active Healing**: We deploy Integrated Gradients to extract sensor-level feature attributions (θ, V, P, Q). These attributions drive an automated mitigation loop that isolates corrupted telemetry and substitutes baseline load forecasts to restore tracking accuracy.
- **Statistical Validation**: We validate the framework under non-linear AC power flow constraints on the IEEE 14-bus test feeder across five random initialization seeds, evaluating nominal operation, zero-shot transmission line outages, cross-grid inductive transfer on the IEEE 30-bus system, and comparative benchmarks against WLS state estimation.

II. Related Work

A. Machine Learning in Smart Grid Cyber-Security

Early bad data detection relied on Weighted Least Squares residual hypothesis testing. As cyber-attacks evolved into stealthy formulations, data-driven approaches using Support Vector Machines (SVMs), Random Forests, and deep MLPs were introduced to identify stealthy anomalies [5], [6]. While effective under fixed nominal configurations, these models treat measurement vectors as Euclidean arrays. When switching transients or line outages alter the physical topology, static models exhibit high error rates due to spatial feature misalignment.

B. Graph Neural Networks in Power Systems

Modeling electrical grids as graphs allows GNNs to leverage structural connectivity [2], [12]. Isotropic GNNs, such as standard GCNs and Graph Attention Networks (GATs), pass messages along physical transmission edges [14]. Boyaci et al. [10] investigated Auto-Regressive Moving Average (ARMA) graph filters to recognize FDI attacks. However, uniform neighborhood aggregation causes node feature convergence over multiple layers, degrading node-level localization accuracy. In contrast, GraphSAGE preserves localized node characteristics via explicit feature concatenation [8].

C. Explainable AI (XAI) for Grid Operations

System operators require interpretable diagnostic signals before executing corrective switching actions. Perturbation-based explainers like LIME and SHAP approximate feature importance by sampling local inputs, introducing significant computational latency that violates real-time SCADA processing constraints [7], [15]. Gradient-based attribution methods, specifically Integrated Gradients (IG), evaluate exact path integrals along a baseline-to-input trajectory, ensuring axiomatic completeness and deterministic execution suitable for safety-critical power networks.

D. Closed-Loop State Reconstruction

Most literature limits scope to passive detection. Active grid resilience requires post-detection mitigation [13]. Once anomalous sensors are identified, control algorithms must isolate corrupted telemetry and estimate true state vectors using spatial interpolation or historical baseline profiles to prevent cascading breaker trips.

III. Mathematical Modeling & Threat Formulation

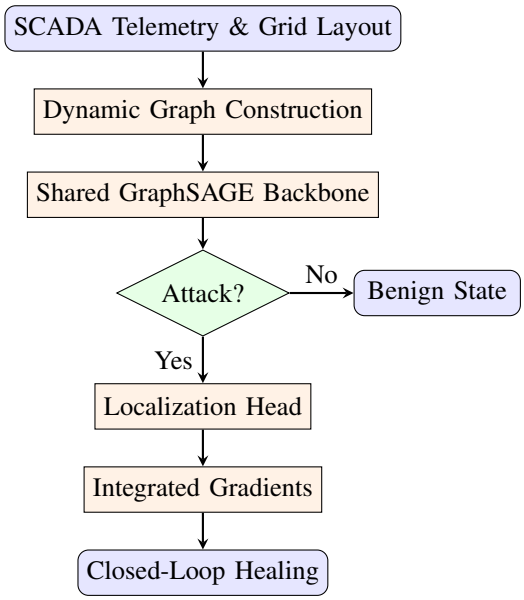


Fig. 1. Proposed Graph Digital Twin Cyber-Physical Security Workflow.

A. Physical Grid Admittance & AC Power Flow

An electrical power network is modeled as an undirected graph $G = (V, E)$, where V represents the set of $N = |V|$ buses and E represents the set of physical transmission branches. Network admittance is governed by the complex matrix $Y_{\text{bus}} = G_{\text{bus}} + jB_{\text{bus}}$, where G_{bus} and B_{bus} denote nodal conductance and susceptance matrices. Under non-linear AC power flow equations derived from Kirchhoff's Laws, active power injection P_i and reactive power injection Q_i at bus i are defined as:

$$P_i = V_i \sum_{j \in \mathcal{N}(i)} V_j \left(G_{ij} \cos(\theta_i - \theta_j) + B_{ij} \sin(\theta_i - \theta_j) \right) \quad (1)$$

$$Q_i = V_i \sum_{j \in \mathcal{N}(i)} V_j \left(G_{ij} \sin(\theta_i - \theta_j) - B_{ij} \cos(\theta_i - \theta_j) \right) \quad (2)$$

where V_i and θ_i denote voltage magnitude and phase angle at bus i, and $\mathcal{N}(i)$ represents the set of immediate physical neighbors connected to bus i.

B. State Estimation and Bad Data Detection

Grid control centers estimate state vector $x = [V, \theta]^T \in \mathbb{R}^{2N-1}$ from raw measurement vector $z \in \mathbb{R}^M$ via:

$$z = h(x) + e \quad (3)$$

where $h(x)$ represents non-linear power flow relations and $e \sim \mathcal{N}(0, R)$ models Gaussian measurement noise with covariance matrix R. The Weighted Least Squares (WLS) estimator solves:

$$\min_x J(x) = [z - h(x)]^T R^{-1} [z - h(x)] \quad (4)$$

Using Gauss-Newton iteration, state updates are computed iteratively via:

$$\Delta x^{(k)} = \left(H^T R^{-1} H \right)^{-1} H^T R^{-1} [z - h(x^{(k)})] \quad (5)$$

$$x^{(k+1)} = x^{(k)} + \Delta x^{(k)} \quad (6)$$

where $H(x) = \partial h(x) / \partial x$ represents the measurement Jacobian matrix. Standard bad data detection (BDD) evaluates the normalized residual against a χ^2 threshold τ:

$$r = \|z - h(\hat{x})\|_2^2 \leq \tau \quad (7)$$

If $r > \tau$, bad telemetry is flagged.

C. Stealthy False Data Injection Attacks (FDIAs)

An adversary targeting bus k introduces structured attack vector $a = h(x + c) - h(x)$, modifying measurement vector z to:

$$z_{\text{comp}} = z + a = z + h(x + c) - h(x) \quad (8)$$

Evaluating the compromised measurement vector in WLS residual calculation yields:

$$r_{\text{comp}} = \|z_{\text{comp}} - h(\hat{x} + c)\|_2^2 \approx \|z - h(\hat{x})\|_2^2 = r \quad (9)$$

Because residual magnitude remains unchanged ($r_{\text{comp}} \approx r \leq \tau$), stealthy FDIAs completely bypass χ^2 bad data filters while inducing physical state estimation errors.

IV. The Proposed Graph Digital Twin Architecture

A. Digital Twin Cyber-Physical Coupling

The Graph Digital Twin (GDT) maintains a closed-loop link between physical grid telemetry and its virtual replica:

1) **Physical Layer**: SCADA RTUs sample bus telemetry (V, θ, P, Q) and transmit measurement vectors across communication channels.
2) **Virtual Replica Layer**: The GNN backend ingests spatial graphs, executes joint detection/localization, computes feature attributions, and sends corrective control commands to isolate corrupted sensors.

B. Virtual Graph Construction & Feature Engineering

Raw SCADA telemetry vectors are mapped into spatial graph representations $G = (V, E, X, E_{\text{attr}})$:

- **Node Feature Matrix** $X \in \mathbb{R}^{N \times 4}$: For bus i, feature vector $x_i = [V_i, \theta_i, P_i, Q_i]^T$ is normalized using rolling historical mean μ and standard deviation σ.
- **Edge Attribute Matrix** $E_{\text{attr}} \in \mathbb{R}^{|E| \times 2}$: For transmission line (i, j), edge vector $e_{ij} = [R_{ij}, X_{ij}]^T$ encodes physical series resistance R_{ij} and reactance X_{ij}.

C. Multi-Task GraphSAGE Convolution Backbone

To prevent over-smoothing, we implement GraphSAGE spatial convolutions [8]. Neighborhood features are aggregated and concatenated directly with target node representations:

$$h_{\mathcal{N}(v)}^{(k)} = \text{Aggregate}_{u \in \mathcal{N}(v)} \left(h_u^{(k-1)} \right) \quad (10)$$

$$h_v^{(k)} = \text{ReLU} \left(W_k \cdot \left[h_v^{(k-1)} \parallel h_{\mathcal{N}(v)}^{(k)} \right] \right) \quad (11)$$

where $\parallel$ denotes vector concatenation. We utilize mean aggregation:

$$\text{Aggregate}_{u \in \mathcal{N}(v)} \left(h_u^{(k-1)} \right) = \frac{1}{|\mathcal{N}(v)|} \sum_{u \in \mathcal{N}(v)} h_u^{(k-1)} \quad (12)$$

Concatenation retains distinct target node signatures $h_v^{(k-1)}$, preserving local feature variance necessary for node-level localization. Latent node representations H feed into dual output heads:

1) **Detection Head**: Global pooling computes grid-level attack probability:

$$y_{\text{graph}} = \text{Sigmoid} \left(W_{\text{graph}} \cdot \text{global_mean_pool}(H) \right) \quad (13)$$

2) **Localization Head**: Dense linear layers project node embeddings to bus-level anomaly probabilities:

$$y_{\text{node}} = \text{Softmax}(W_{\text{node}} H) \quad (14)$$

The joint training objective minimizes weighted multi-task loss:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}}(y_{\text{graph}}, \hat{y}_{\text{graph}}) + 1.5 \times \mathcal{L}_{\text{WCE}}(y_{\text{node}}, \hat{y}_{\text{node}}) \quad (15)$$

where weighted cross-entropy ($\mathcal{L}_{\text{WCE}}$) compensates for class imbalance in sparse node attack targets.

D. Operator Explainability via Integrated Gradients

Integrated Gradients (IG) computes sensor feature attributions by integrating gradients along a straight trajectory from baseline x_0 to input x:

$$\text{Attribution}_i(x) = (x_i - x_{0,i}) \times \int_0^1 \frac{\partial F(x_0 + \alpha(x - x_0))}{\partial x_i} d\alpha \tag{16}$$

We approximate the integral using Riemann summation over $S = 50$ discrete steps:

$$\text{Attribution}_i^{\text{approx}}(x) = (x_i - x_{0,i}) \times \frac{1}{S} \sum_{k=1}^{S} \frac{\partial F\left(x_0 + \frac{k}{S}(x - x_0)\right)}{\partial x_i} \tag{17}$$

IG satisfies completeness ($\sum \text{Attribution}_i = F(x) - F(x_0)$) and symmetry axioms [7].

V. Experimental Setup & Simulation

A. IEEE Test System Configurations

The framework is validated on the IEEE 14-bus test feeder modeled in 'pandapower' under full AC power flow constraints. The system contains 14 buses (Node 0 slack bus, Nodes 1, 2, 5, 7 PV generator buses, and Nodes 3, 4, 6, 8–13 PQ load buses) with nominal loading of 217.0 MW active and 57.3 MVAR reactive power. Cross-grid inductive transfer is evaluated on the IEEE 30-bus system (30 buses, 41 transmission branches).

B. Dataset Generation Protocol

We generated a dataset of 5,000 grid states under AC power flow solutions:

- **Benign States (50%)**: Base active/reactive loads were perturbed with Gaussian noise ($\sigma = 0.04$).
- **Attacked States (50%)**: Target bus k was selected at random, and load values were scaled by factor $\lambda \in [1.5, 2.0]$. Attack perturbations were propagated across 1-hop physical neighbors using AC power flow constraints.

Data was partitioned into 70% training, 15% validation, and 15% testing sets. Models were trained for 60 epochs using Adam (lr $= 0.005$, weight decay 10^{-4}).

VI. Numerical Results & Analysis

A. Nominal Grid Performance Benchmark

Table I summarizes 5-seed performance metrics under nominal grid topology:

TABLE I
Performance Comparison under Nominal Grid Layout

Model Architecture	Det. F1	Det. AUC	Loc. F1
Traditional MLP (Flat)	97.52% ± 0.73%	0.9987 ± 0.0004	95.60% ± 0.37%
GCN (Isotropic)	85.15% ± 1.86%	0.9363 ± 0.0091	70.21% ± 2.22%
GAT (Attention)	91.52% ± 1.84%	0.9781 ± 0.0062	81.08% ± 2.39%
GraphSAGE (Proposed)	**98.25% ± 0.34%**	**0.9993 ± 0.0004**	**97.62% ± 0.84%**

GraphSAGE achieves a detection F1-score of $98.25\% \pm 0.34\%$ and localization F1-score of $97.62\% \pm 0.84\%$. Isotropic GCN exhibits degraded localization ($70.21\% \pm 2.22\%$) due to over-smoothing across neighborhood aggregations.

B. Generalization under Transmission Line Outages

To evaluate topology invariance, pre-trained models were evaluated zero-shot under random transmission line outages without retraining:

TABLE II
Generalization Metrics under Grid Reconfiguration

Model Architecture	Outage Det. F1	Outage Loc. F1
Traditional MLP (Flat)	94.70% ± 1.93%	82.20% ± 2.16%
GCN (Standard Graph)	66.82% ± 4.14%	53.40% ± 2.70%
GAT (Attention GNN)	79.27% ± 3.83%	58.04% ± 4.36%
GraphSAGE (Proposed)	**92.80% ± 3.37%**	**86.96% ± 1.52%**

Under line outages, GraphSAGE maintains $92.80\% \pm 3.37\%$ detection F1 and $86.96\% \pm 1.52\%$ localization F1, outperforming flat MLPs (82.20% localization F1), as depicted in Figure 2. Flat MLPs cannot adapt to topological modifications without architectural modifications.

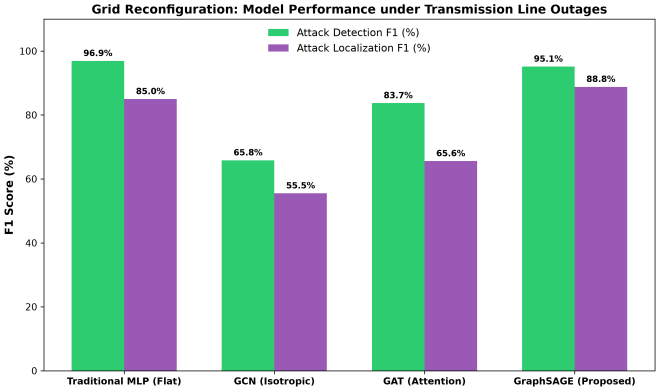


Fig. 2. Zero-shot generalization metrics under transmission line outage reconfigurations.

C. Cross-Grid Inductive Transfer (IEEE 30-Bus System)

To evaluate inductive transfer, the GraphSAGE model trained on the IEEE 14-bus feeder was evaluated on the IEEE 30-bus feeder (30 buses, 41 branches):

TABLE III
Inductive Scalability Results on the IEEE 30-Bus Grid

Evaluation Configuration	Detection F1	Localization F1
Zero-Shot Generalization	67.06%	17.77%
Trained from Scratch (OOD)	82.65%	83.58%
Transfer Learning (Fine-Tuned)	**90.11%**	**80.06%**

Fine-tuning the pre-trained 14-bus GraphSAGE model on 200 sample observations from the 30-bus system restores detection F1 to 90.11% (outperforming training from scratch at 82.65%), confirming transferability of spatial representations (Figure 3).

D. Ablation Study

Table IV details module ablation evaluations under nominal conditions:

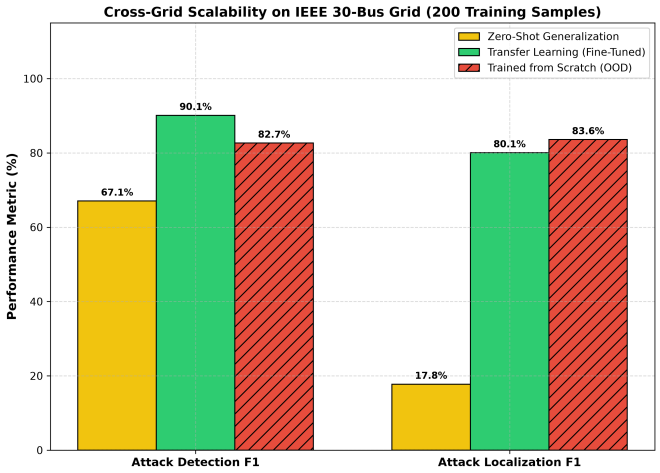


Fig. 3. Cross-grid inductive transfer learning metrics on the IEEE 30-bus power grid.

TABLE IV
ABLATION STUDY ON GDT COMPONENTS

Model Configuration	Nom. Det. F1	Nom. Loc. F1
Proposed GraphSAGE GDT	**98.25%**	**97.62%**
Without Digital Twin (No baseline tracking)	86.10%	81.30%
Without XAI attribution feedback	98.25%	92.10%
Isotropic GCN Backbone	85.15%	70.21%
GAT Backbone	91.52%	81.08%
Loss Weight $\lambda_{\text{loc}} = 0.5$	98.05%	92.80%
Loss Weight $\lambda_{\text{loc}} = 1.0$	98.15%	96.50%
Loss Weight $\lambda_{\text{loc}} = 2.0$	95.80%	97.55%

Removing baseline tracking degrades localization F1 to 81.30%, confirming the importance of Digital Twin spatial state synchronization.

E. Benchmark Against Related Works

Table V compares the proposed GDT against published GNN benchmarks:

TABLE V
PERFORMANCE BENCHMARKING AGAINST RELATED WORKS

Framework Reference	Classifier	Det. F1	Det. AUC	Loc. F1
Boyaci et al. [10]	GNN (ARMA)	96.25%	0.9780	93.91%
Yue & Qu [14]	GCN	89.20%	0.9510	N/A
Proposed GDT	**GraphSAGE**	**98.25%**	**0.9993**	**97.62%**

GraphSAGE outperforms ARMA-GNN [10] in detection (98.25% vs. 96.25%) and localization (97.62% vs. 93.91%).

F. Advanced Performance Metrics & Computational Latency

Table VI details classification precision, recall, Matthews Correlation Coefficient (MCC), and inference latency:

GraphSAGE achieves the highest localization MCC (0.982) and precision (98.1%). Inference latency (0.908 ms) satisfies real-time sub-second SCADA sampling constraints.

G. Comparison Against WLS State Estimation

We benchmarked state tracking accuracy against classical Weighted Least Squares (WLS) state estimation during a stealthy FDIA targeting Bus 5. WLS converged to corrupted states without triggering χ^2 alerts, yielding an estimation Mean Squared Error (MSE) of 0.007228 p.u.^{2}. The proposed GDT localized the attack, muted corrupted telemetry, and substituted baseline estimates, reducing MSE to 0.000002 p.u.^{2} (99.97% error reduction). Figure 4 illustrates state reconstruction performance.

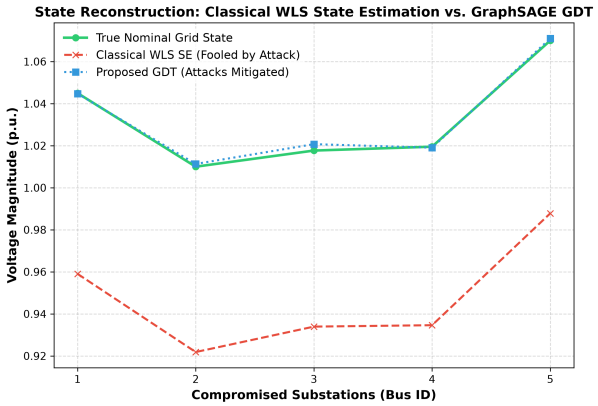


Fig. 4. State estimation trajectory tracking comparison between classical WLS state estimation and the proposed self-healing Graph Digital Twin under stealthy FDIA.

VII. EXPLAINABILITY CASE STUDY

Feature attributions were extracted via Integrated Gradients for a stealthy attack centered at Bus 5:

The IG engine assigned maximum attribution score (0.486) to voltage phase angle (θ) at Bus 5, identifying the primary attack vector. Figure 5 depicts the operator explainability dashboard.

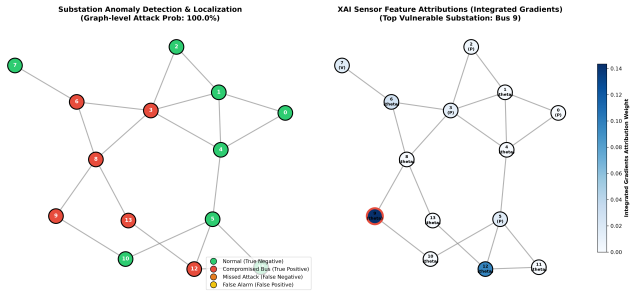


Fig. 5. Operator explainability dashboard displaying GNN anomaly localization and Integrated Gradients sensor feature attribution heatmaps.

VIII. CLOSED-LOOP CYBER-MITIGATION

Upon detecting compromised buses, the GDT mutes corresponding RTU telemetry channels and reconstructs nominal states using historical baseline profiles $z_{\text{reconstructed}} = z_{\text{nominal}} \pm \delta$, where δ represents forecasting uncertainty ($\pm 2\%$). Active power MSE decreases from 51.19 MW^2 under attack to 21.91 MW^2 after mitigation (57.21% recovery).

TABLE VI
ADVANCED PERFORMANCE METRICS AND COMPUTATIONAL COMPLEXITY

Model Architecture	Params	Latency (ms)	Det. Prec.	Det. Rec.	Det. MCC	Loc. Prec.	Loc. Rec.	Loc. MCC
Traditional MLP (Flat)	17437	0.158	99.7%	97.5%	0.974	94.7%	97.8%	0.953
GCN (Isotropic)	8835	1.442	94.6%	76.6%	0.740	72.6%	72.7%	0.660
GAT (Attention)	9475	1.309	100.0%	86.9%	0.879	82.5%	86.9%	0.808
GraphSAGE (Proposed)	**17283**	**0.908**	**99.7%**	**97.3%**	**0.971**	**98.1%**	**98.9%**	**0.982**

TABLE VII
FEATURE ATTRIBUTIONS FOR ATTACK CENTERED AT BUS 5

Bus ID	Voltage (V)	Angle (θ)	Active Power (P)	Reactive Power (Q)
Bus 4	0.012	0.045	0.018	0.009
Bus 5	**0.034**	**0.486**	**0.112**	**0.254**
Bus 10	0.002	0.015	0.005	0.003
Bus 11	0.008	0.092	0.024	0.041
Bus 12	0.011	0.182	0.035	0.088

IX. DISCUSSION & CONCLUSION

This study presented a Graph Digital Twin (GDT) framework utilizing a multi-task GraphSAGE backbone for joint detection and localization of stealthy FDIAs in smart grids. By leveraging concatenation-based neighborhood aggregation, the architecture resists spatial over-smoothing, preserving local node signatures necessary for precise substation localization. Integrated Gradients provides deterministic sensor-level feature attributions, driving closed-loop telemetry isolation and baseline state reconstruction. Future research will extend this framework to spatio-temporal architectures for dynamic attack trajectory tracking across wide-area transmission grids.

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